

EXPERIMENT NO: 18

GENERATION OF PWM USING TIMER INTERRUPT

AIM:

To generate PWM using Timer interrupt.

ECA14 Embedded Systems Lab Manu...

APPARATUS / SOFTWARE REQUIRED:

- PC with Windows OS
- Keil μ Vision IDE
- 8051 Simulator (Keil built-in)
- Serial Window (UART #1)

PROGRAM:

```
#include <reg51.h>
```

```
sbit PWM = P1^0;
```

```
unsigned char count = 0;
```

```
unsigned char duty = 100; // Change this value (0–200)
```

```
void timer0_ISR(void) interrupt 1
```

```
{
```

```
    TH0 = 0xFC;           // Reload for 1 ms delay
```

```
    TL0 = 0x66;
```

```
    count++;
```

```
    if(count < duty)
```

```
        PWM = 1;         // ON time
```

```
    else
```

```

    PWM = 0;           // OFF time

    if(count >= 200)    // Total period = 200 ms

        count = 0;

}

void main(){

    TMOD = 0x01;        // Timer0 Mode 1 (16-bit)

    TH0 = 0xFC;         // 1 ms preload

    TL0 = 0x66;

    IE = 0x82;          // EA = 1, ET0 = 1

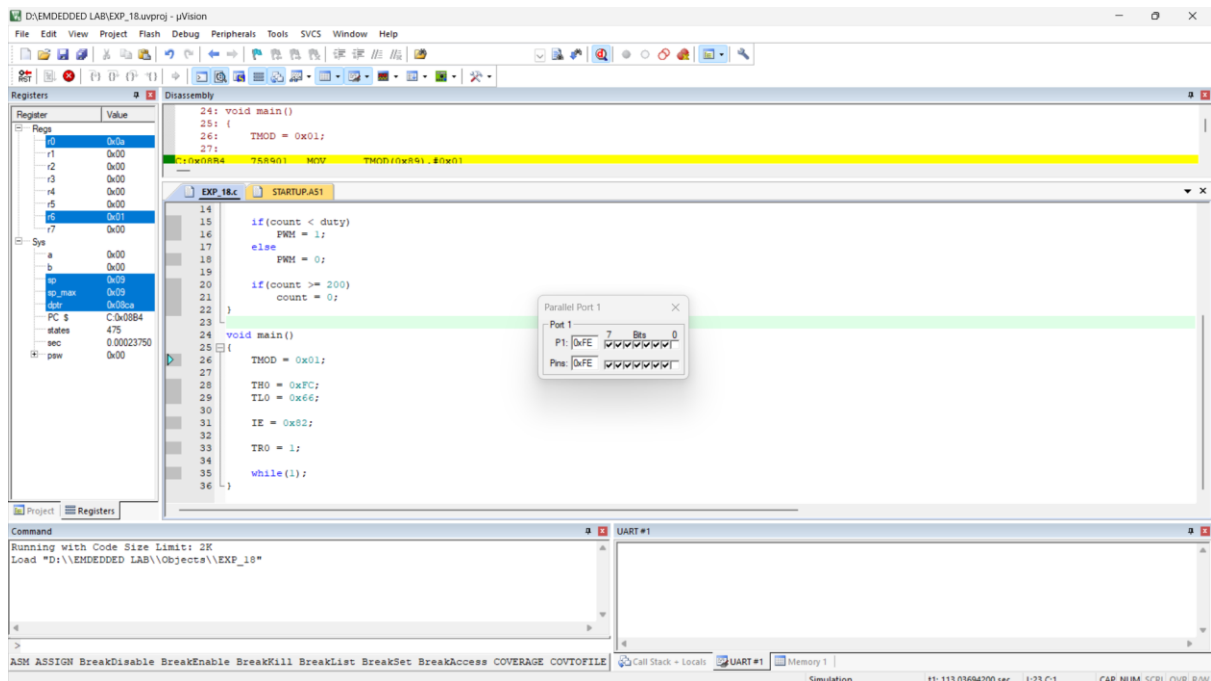
    TR0 = 1;            // Start Timer0

    while(1);

}

```

OUTPUT:



FORMULA:

$$\text{Duty Cycle (\%)} = (\text{ON time} / \text{Total period}) \times 100$$

RESULT:

Thus the PWM was generated successfully using **Timer Interrupt** for various duty cycles.